# Nikolas Melissaris

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## Employment

**IRIF, CNRS & Université Paris-Cité** · *Paris, France* 2025–Present
Postdoctoral Researcher, *hosted by Geoffroy Couteau*

## Education

**Aarhus University** · *Aarhus, Denmark* 2025
PhD in Computer Science
Thesis: Better, Faster, Stronger: Improving Security, Efficiency, and Primitives for MPC [pdf]
*Advisors: Peter Scholl, Claudio Orlandi*

**Rutgers – The State University of New Jersey** · *New Jersey, USA* 2021
MSc in Information Technology

**National Technical University of Athens** · *Athens, Greece* 2015
BSc & MSc in Applied Mathematics
Majors: Discrete Mathematics, Probability and Statistics

## Research Experience

**Institut de Recherche en Informatique Fondamentale** · *Paris, France* Spring '24
Research Visit, *hosted by Geoffroy Couteau*

**JP Morgan, AlgoCRYPT Group** · *New York City, USA* Summer '23
Research Intern – Privacy-preserving ML

**Capital Fund Management** · *New York City, USA* Summer '21
Research Intern – Clustering methods for financial time series

**MadHive Inc.** · *New York City, USA* Summer '19
Research Assistant – Cryptographic integrity for AdTech

**UC Santa Barbara, Computer Security Lab** · *Santa Barbara, USA* Summer '15
Research Assistant – Android GPS spoofing defense

# Teaching

## Instructor

| Institution | Term | Hours | Course · Students |
|---|---|---|---|
| ESILV Paris | Fall '25 | 8 | Zero-Knowledge Proofs [link] · 12 |
| Rutgers University | Summer '20 | 40 | Management Information Systems · 30 |
| Columbia University | Summer '17 | 55 | Introduction to Programming with C · 25 |
| Borough of Manhattan Community College | Spring '17 | 38 | Principles in Information Science and Computing [link] · 20 |
| NYC College of Technology | Spring '17 | 38 | Quantitative Reasoning [link] · 25 |
| | Fall '16 | 50 | Discrete Structures and Algorithms I [link] · 20 |
| Brooklyn College | Fall '16 | 38 | Intro to Computer Applications [link] · 30 |
| | Spring '17 | 38 | |

## Teaching Assistant

| Institution | Term | Hours | Course · Students |
|---|---|---|---|
| Aarhus University | Fall '23 | 15 | Cryptology · 25 |
| | Fall '22 | 15 | |
| | Spring '23 | 5 | Computability and Logic · 18 |
| | Spring '22 | 40 | Optimization · 9 |
| Rutgers University | Spring '21 | 15 | Information Security · 30 |
| | Fall '20 | 15 | |
| | Fall '18 | 15 | Database Systems · 60 |
| | Spring '18 | 15 | Business Data Management [link] · 120 |
| | Spring '19 | 15 | |
| | Fall '17 | 10 | Optimization · 30 |
| | Fall '17 | 10 | Statistics · 30 |

*Note: The distinction between the two tables lies in the level of responsibility. As an instructor I was solely responsible for the course, from setting the curriculum, to teaching, to office hours, to grading. As a teaching assistant the workload revolved around exercise sessions, office hours, and grading. The hours mentioned are only in-class hours, per semester, either lecturing or for exercise sessions, and do not include preparation and grading. The number of students mentioned is per semester and it refers to the group of students that I was responsible for, not (always) the whole class. I've linked to course materials where I still have them and they're mine to share. Some are lost to time and others are stuck behind institutional access I no longer have.*

## Academic Service

### Subreviewer

| | |
|---|---|
| **CRYPTO** | (2021, 2025, 2026) |
| **EUROCRYPT** | (2022, 2024, 2025, 2026) |
| **ASIACRYPT** | (2024, 2025) |
| **TCC** | (2019, 2023) |
| **PKC** | (2026) |

## Papers

### Published

4. *Benny Applebaum, Dung Bui, Geoffroy Couteau, and Nikolas Melissaris.* Structured-Seed Local Pseudorandom Generators and their Applications. **APPROX/RANDOM 2025** [pdf]
3. *Carsten Baum, Nikolas Melissaris, Rahul Rachuri, and Peter Scholl.* Cheater Identification on a Budget: MPC with Identifiable Abort from Pairwise MACs. **CRYPTO 2024** [pdf]
2. *Nikolas Melissaris, Divya Ravi, and Sophia Yakoubov.* Threshold-optimal MPC with Friends and Foes. **INDOCRYPT 2023** [pdf]
1. *Pei Peng, Nikolas Melissaris, Emina Soljanin, Bill Lee, Anton Maliev, and Huafeng Fan.* Straggling for Covert Message Passing on Complete Graphs. **Allerton 2019** [pdf]*

### Manuscripts

4. *Diego F. Aranha and Nikolas Melissaris.* Scanning the Social Contract: Freedom, Fear, and the Limits of Technological Obedience. In Submission.
3. *Diego F. Aranha and Nikolas Melissaris.* What is Cryptography Hiding from Itself?. **Real World Crypto 2026** · **Undone Computer Science 2026** [pdf]
2. *Geoffroy Couteau, Alexandrer Koch, Nikolas Melissaris, Peter Scholl, Sacha Servan-Schreiber, and Xiaxi Ye.* On Compressing Non-Additive Correlations. Cryptology ePrint Archive. [pdf]
1. *Nikolas Melissaris, Jiayi Xu, Antigoni Polychroniadou, Akira Takahashi, and Chenkai Weng.* ZKBoost: Zero-Knowledge Verifiable Training for XGBoost. Cryptology ePrint Archive. [pdf]*

*Note: In cryptography (and theoretical computer science in general) the author list is alphabetical. Research in this field tends to happen in small groups of people. It becomes hard to say whose idea contributed more to what. Alphabetical ordering is a clean way to acknowledge that and default to "we all contributed," which typically reflects reality reasonably well. Most publications above follow this convention, except the ones marked with an asterisk, which are published in different venues where contribution-based ordering is standard.*

## Talks

### Conferences & Workshops

| | | |
|---|---|---|
| March 2026 | Undone Computer Science | · *Luxembourg, Luxembourg* |
| March 2026 | Real World Crypto [video] | · *Taipei, Taiwan* |
| August 2024 | CRYPTO [video] | · *Santa Barbara, USA* |
| June 2024 | Theory and Practice of MPC [video] | · *Darmstadt, Germany* |

### Invited Talks

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|---|---|
| April 2026 | INFRES seminar Télécom · [slides] · *Palaiseau, France* |
| March 2026 | IRIF non-permanents' seminar · [slides] · *Paris, France* |
| February 2026 | École Polytechnique · [slides] · *Palaiseau, France* |
| December 2025 | Séminaire Algorithmique GREYC · [slides] · [page] · *Caen, France* |
| December 2025 | Crypto Seminar CAPSULE · [slides] · [page] · *Rennes, France* |
| November 2025 | Seminar INSPIRE · [slides] · *Saclay, France* |
| October 2025 | Crypto Seminar LIRMM · [slides] · [page] · *Montpellier, France* |
| February 2024 | AlgoComp Seminar IRIF · [slides] · [page] · *Paris, France* |
| September 2023 | AI Research JP Morgan [slides] · *New York City, USA* |
| December 2022 | AlgoCRYPT Seminar JP Morgan · *New York City, USA* |
| June 2022 | Crypto Summer Day [slides] · *Aarhus, Denmark* |

### Science Outreach

| | |
|---|---|
| March 2022 | AU Hack [slides] · *Aarhus, Denmark* |

*Note: Where available, links to videos and slides are included. For invited talks, some research groups maintain public seminar pages with speaker announcements, which I've linked. Others don't publicize their seminars online or the pages are no longer available.*

## Awards and Honors

**Stibofonden** · 50,000 dkk · 2024
**Rutgers Summer Research** · 3,000 usd · 2019, 2020

*Note: The Stibofonden grant is awarded to PhD students for extended research visits abroad. The Rutgers Summer Research Scholarship is a merit-based award granted on the basis of a submitted research proposal.*